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Technical Communications for Engineers and Computer Scientists ✓

Ms. Alysoun Taylor-Hall ✓

Recommendation Report ✓

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Selecting the Best Password Manager for College Students ✓

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Introduction

Purpose

The purpose of this report is to select and recommend the best password manager for college students.

Problem

College students use multiple online services such as email, banking, learning platforms, and social media accounts. Many students reuse passwords or choose weak passwords due to difficulty remembering multiple credentials, creating significant cybersecurity risks, including unauthorized account access and identity theft. Research shows that password manager interaction problems with websites reduce adoption and lead users to rely on insecure workarounds [1]. A password manager improves security by storing and generating strong, unique passwords. Students require a solution that is affordable, secure, and easy to use across multiple devices.

Three password managers are being considered: Bitwarden, 1Password, and LastPass.

Scope

This report compares Bitwarden, 1Password, and LastPass based on security features, ease of use, and annual cost. The report focuses on plans appropriate for individual student use.

Discussion

Security Features

Explanation

Security evaluation refers to the ability of each password manager to protect stored credentials from unauthorized access. Key factors include encryption standards, zero-knowledge architecture, multi-factor authentication (MFA) support, and security audit history. Research comparing open-source password manager applications using ISO/IEC 25010 quality standards shows meaningful differences in how password managers protect user data [2]. The strength of a password manager depends on its cryptographic implementation, including AES-256 symmetric encryption, which is widely considered resistant to brute-force attacks [3]. ✓

Data

Bitwarden uses AES-256 encryption, zero-knowledge architecture, and supports MFA including TOTP and YubiKey hardware keys [4]. Its open-source codebase is publicly available for independent security audits, and no major breaches have been reported. 1Password uses AES-256 encryption with an additional Secret Key for vault decryption and holds SOC 2 Type 2 certification [5]. LastPass uses AES-256 encryption and MFA but suffered a major data breach in 2022 in which encrypted vault data was exfiltrated by attackers [6]. Table 1 compares the security features of each password manager.

Feature	Bitwarden	1Password	LastPass
Encryption	AES-256 [4]	AES-256 [5]	AES-256 [6]
Architecture	Zero-knowledge [4]	Zero-knowledge + Secret Key [5]	Zero-knowledge [6]
MFA Support	TOTP, email, YubiKey [4]	Yes [5]	Yes [6]
Open Source	Yes [4]	No [5]	No [6]
Breach History	None reported [4]	None reported [5]	Yes - 2022 [6]
Security Rank	2	1	3

 ✓

Source: Author

Interpretation

Bitwarden and 1Password both offer strong security architectures. Bitwarden's open-source model provides additional transparency and community oversight. 1Password's Secret Key provides a meaningful additional protection layer. LastPass ranks third due to the documented 2022 breach. Overall, 1Password ranks first for security, followed by Bitwarden, then LastPass.

Ease of Use

Good

Explanation

Ease of use evaluates how easily students can install, configure, and use the password manager in their daily workflow. Key factors include browser integration, mobile support, cross-platform compatibility, and user interface quality. Research demonstrates that interaction problems between password managers and websites significantly reduce adoption rates among users [1].

Data

Bitwarden offers browser extensions for all major browsers, mobile apps for iOS and Android, and desktop clients for Windows, macOS, and Linux [4]. 1Password is widely recognized for its polished interface, browser extensions, mobile and desktop clients, and passkey storage support [5]. LastPass provides strong browser integration but has received mixed usability reviews following platform changes, including removal of free-tier multi-device support in 2021 [6].

Interpretation

1Password provides the best overall user experience due to its polished interface. Bitwarden offers strong usability with full cross-platform compatibility. LastPass falls behind due to recent platform restrictions. 1Password ranks first, Bitwarden second, and LastPass third in usability.

Annual Cost

Good

Explanation

Cost evaluates the annual subscription price for individual students. For students with limited budgets, affordability is a significant factor in adoption. A solution that requires no payment for core functionality lowers the barrier to entry.

Data

Bitwarden offers a free plan with unlimited password storage across unlimited devices and a premium plan costing approximately \$19.80 per year [4]. 1Password does not offer a free plan; its individual plan costs approximately \$35.88 per year [5]. LastPass offers a limited free plan restricted to one device type; its premium plan costs approximately \$36 per year [6]. Table 2 compares the annual cost of each password manager.

Password Manager	Free Plan	Annual Premium Cost	Cost Rank
Bitwarden [4]	Yes (unlimited devices)	\$19.80/year	1
1Password [5]	No	\$35.88/year	2
LastPass [6]	Yes (limited —one device type)	\$36/year	3

Source: Author

good

Interpretation

Bitwarden provides the lowest cost, offering both a capable free tier and an affordable premium upgrade. 1Password and LastPass are significantly more expensive annually. For cost-conscious students, Bitwarden is the clear winner in this criterion.

Conclusion

Summary

This report evaluated Bitwarden, 1Password, and LastPass for college student use based on security features, ease of use, and annual cost. 1Password ranked highest in security and

usability, while Bitwarden offered the strongest overall value. Table 3 summarizes the overall comparison across all three criteria.

	Security	Ease of Use	Cost
Bitwarden	2	2	1
1Password	1	1	2
LastPass	3	3	3

Source: Author; 1 = Best, 2 = Middle, 3 = Least

Conclusions

All three password managers provide AES-256 encryption and MFA support. They differ in security track record, usability quality, and cost. 1Password provides the strongest security and user experience but at the highest cost. Bitwarden delivers comparable security and reliable usability at a significantly lower cost. LastPass is the least recommended option due to its 2022 breach, declining usability reputation, and free-tier device restrictions.

Recommendation

Bitwarden is recommended as the password management solution for college students. It provides strong AES-256 encryption, a zero-knowledge architecture, open-source transparency, and the lowest annual cost: free for basic use or approximately \$19.80 per year for premium access. For students who require the most polished interface and can accommodate a higher budget, 1Password is the preferred alternative.

Contact

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Very nicely done!

References

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Well researched
Good work

